

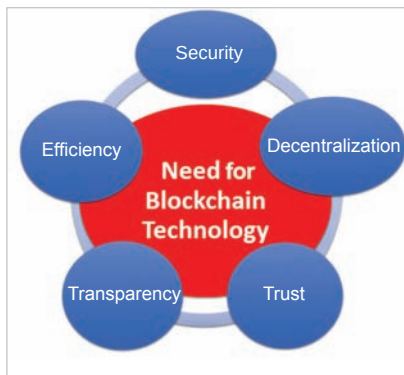


Figure 1: The need for blockchain technology

Healthcare: Blockchain tech can be used to securely store and share medical records. This can improve the accuracy and accessibility of medical data, while also protecting patient privacy.

Real estate: Blockchain technology can be used to create secure and transparent real estate registries. This can help reduce the risk of fraud and increase the efficiency of property transactions.

These are just a few examples of the many potential use cases for blockchain technology. As the technology continues to evolve, we can expect to see it applied to an even wider range of industries and applications.

Why we need blockchain technology

Blockchain technology provides a secure, transparent, and efficient way to store and transmit data, which can help reduce fraud, increase trust, and improve efficiency in various applications.

Security: Blockchain technology provides a high level of security by using advanced cryptographic techniques to protect data. Once data is recorded on a blockchain, it cannot be altered or deleted, making it resistant to fraud and hacking.

Transparency: Blockchain technology provides a transparent and tamper-proof way to record and track data. All participants in a blockchain network can see the same information, and each transaction is recorded in a permanent and unalterable way.

Decentralisation: The blockchain allows for decentralised data storage and transmission, meaning that no single entity has control over the data. This can help reduce the risk of censorship, corruption, and other forms of abuse.

Efficiency: Blockchain tech can help increase efficiency by automating processes and reducing the need for intermediaries. This can help reduce costs, speed up transactions, and increase transparency.

Trust: Blockchain technology can help build trust between parties by providing a secure and transparent way to record and track data. This can be particularly useful in situations where trust is critical, such as financial transactions, supply chain management, and identity verification.

Programming languages and toolkits for blockchain development

There are several programming languages and toolkits that are commonly used in blockchain development. Here are some of the most popular ones.

Solidity: Solidity is a programming language that is used to write smart contracts for the Ethereum blockchain. It is a high-level programming language that is similar to JavaScript and is designed to be easy to learn.

JavaScript: JavaScript is a popular programming language that is used for both front-end and back-end web development. It is often used to build decentralised applications (dApps) that run on blockchain platforms.

Go: Go is a programming language that is frequently used for building blockchain applications. It is known for its simplicity, speed, and concurrency features, making it a good choice for building decentralised applications.

Python: Python is a popular programming language that is used for a wide range of applications, including blockchain development. It is often used for building blockchain applications that require data analysis

and machine learning capabilities.

C++: C++ is a powerful programming language that is used for building high-performance applications, including blockchain applications. It is often used for building blockchain nodes and other backend components.

In addition to these programming languages, there are several toolkits and frameworks that are commonly used in blockchain development. Some of these are listed below.

Ethereum: Ethereum is a popular blockchain platform that is used for building decentralised applications. It provides a variety of tools and frameworks, including the Ethereum Virtual Machine (EVM) and the Solidity programming language.

Hyperledger Fabric: This open source blockchain framework is designed for building enterprise blockchain solutions. It provides a modular architecture that can be customised to meet specific business needs.

Truffle Suite: Truffle Suite is a development framework for Ethereum that provides a suite of tools for building, testing, and deploying smart contracts. It includes the Truffle development environment, the Ganache blockchain emulator, and other useful tools.

Remix: Remix is a web based IDE that is used for developing and testing smart contracts on the Ethereum blockchain. It provides a user-friendly interface and a range of useful features, such as code highlighting, debugging, and testing.

The choice of language and toolkit will depend on the specific requirements of the project, as well as the preferences and expertise of the development team.

Cloud platforms for Blockchain as a Service (BaaS)

Blockchain as a Service (BaaS) is a cloud based service that allows businesses and individuals to develop, host, and use their blockchain